

AI SURGEON

Storyboard — one complete case, panel by panel

This document is the game, told as a story. Thirty-one beats from the moment the patient appears to the moment the player teaches the case to someone else. Every panel carries what the player sees, what it is for, and what has to be built.

SEE ONE · DO ONE · TEACH ONE

The game will not let you cut something you cannot name.

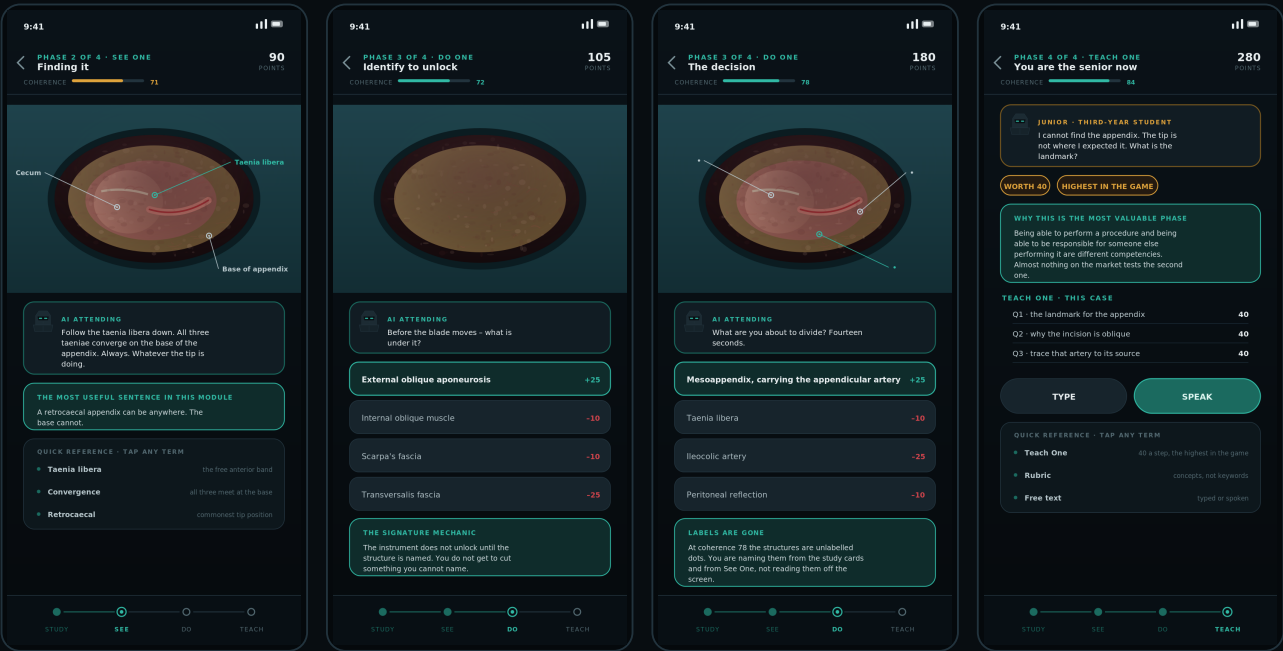
REVISION AUGUST 2026

Phone first.

No headset. No install. It runs in a mobile browser on a device the player already owns. VR is a later phase of the same codebase, not a separate product to raise against.

IN THIS DOCUMENT

Act I · Before you touch anything	beats 1–7
Act II · See One	beats 8–16
Act III · Do One	beats 17–26
Act IV · Teach One	beats 27–31



Screen design is final-intent. Anatomy shown is placeholder — see the note on page 2.

HOW TO READ THIS

Three things on every panel

THE SCREEN

A phone frame showing exactly what the player is looking at during that beat. These are layouts, not concept art. The information hierarchy, the control positions, and the colour language are all intended to be built as drawn.

THE NARRATION

What is happening and why it is in the design. This is the part to read aloud if you are walking someone through the product. It is written so the case makes sense to a reader who has never seen a surgical simulator.

THE BUILD NOTE

The line in small type at the foot of each panel. It says what a developer actually has to implement for that beat — the data structure, the state change, or the rule. Thirty-one of them together are the specification.

WHAT THIS CASE IS

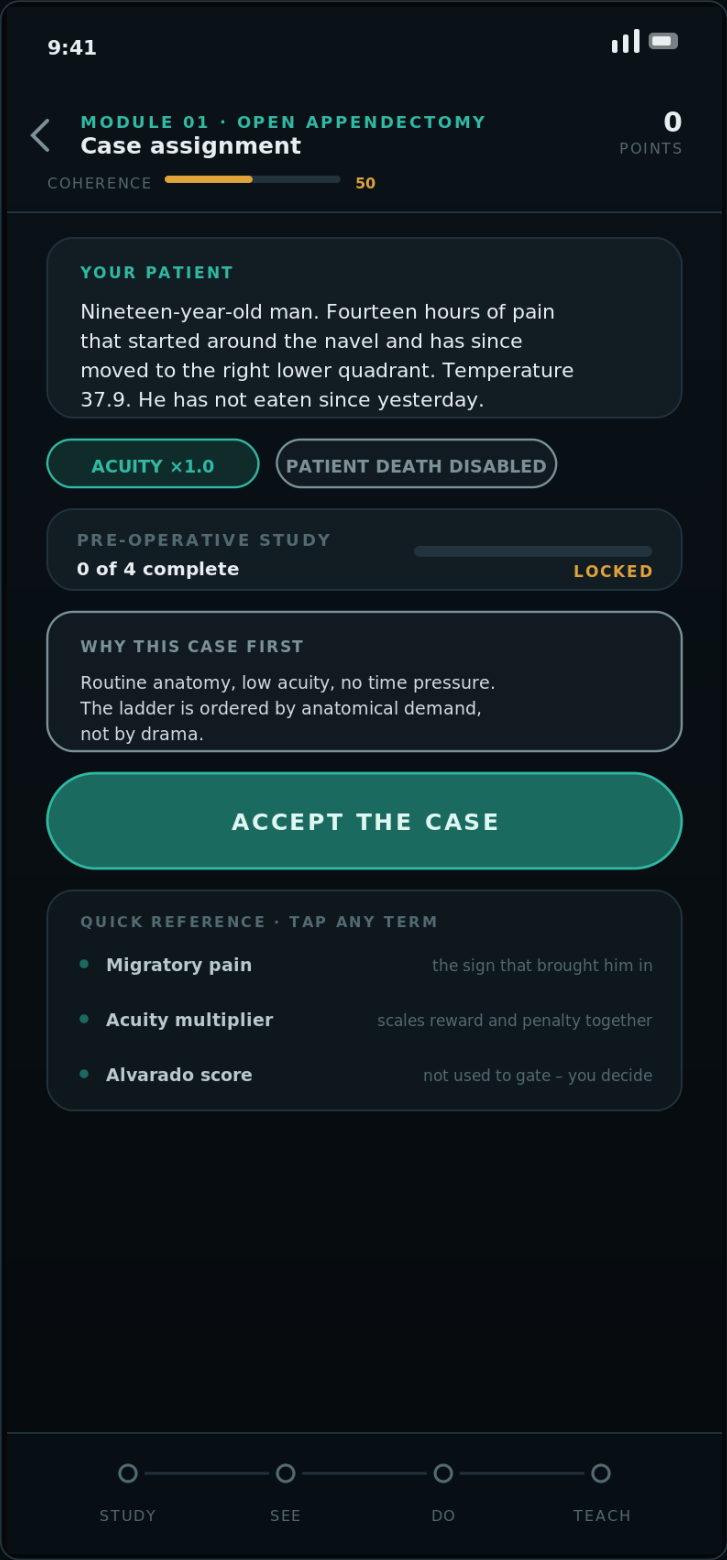
Module 01, open appendectomy through a McBurney gridiron incision. It is the entry module because the anatomy is honest and the acuity is low, not because it looks impressive. A nineteen-year-old with fourteen hours of pain, no complications, patient death disabled. Everything harder on the ladder is the same four phases with a worse patient and a shorter clock.

AN HONEST NOTE ABOUT THE ART

The surgical field in these panels is deliberately abstract. Production anatomy must come from a sourced dataset — BodyParts3D / Z-Anatomy for geometry, the NLM Visible Human cryosections for reference, and real operative video for the See One phase — not modelled from scratch and never built from geometry primitives. Fidelity only has to clear the credibility bar with a surgeon. The engagement comes from the questions, the clock, and the reveal.

THE FOUR PHASES, AND WHAT EACH IS WORTH

STUDY ONE	10 / card	Four cards. A hard gate — the scalpel does not appear until they are done.
SEE ONE	5 / step	The attending operates and names everything. Not skippable, not graded.
DO ONE	25 / step	You operate. Name the structure before the instrument unlocks.
TEACH ONE	40 / step	A junior asks. The most valuable phase, and the untested one.



01 COLD OPEN

The case comes to you

The app does not open on a menu. It opens on a person. Before any mechanic is explained, the player is given a patient with a history that already contains the diagnosis, if they know how to read it.

BUILD A case is a seed: patient demographics, acuity multiplier, and a step list. Everything downstream is generated from it, so two players can be handed the same case and compared.

ON SCREEN, TAPPABLE

- **Migratory pain** the sign that brought him in
- **Acuity multiplier** scales reward and penalty together
- **Alvarado score** not used to gate – you decide



02 THE GATE

You cannot scrub in yet

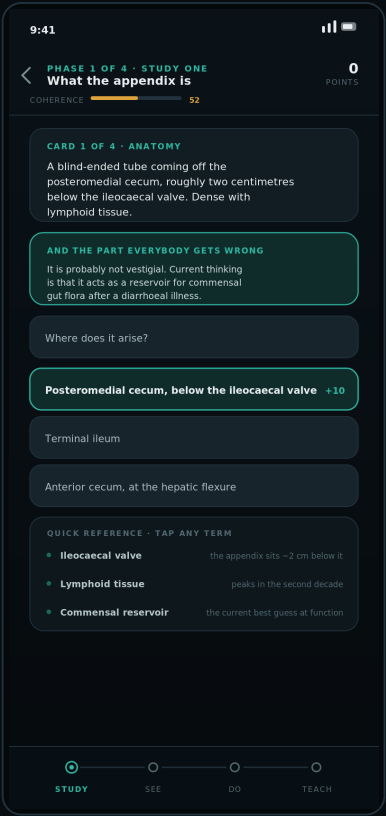
The gate is the whole argument

Every other surgical app on the market opens with an instrument in your hand. This one refuses. The scrub room door does not open until the reading is done, and the player is told exactly why.

BUILD A hard gate, not a nudge. The Do One scene will not instantiate while `case.study.completed < 4`. Ten points a card, banked before the case begins.

ON SCREEN, TAPPABLE

- **Study gate** hard lock, no purchase bypass
- **Coherence** accuracy and hesitation combined
- **Rank** cannot fall in entry modules



03

STUDY ONE

Card 1 · what it actually is

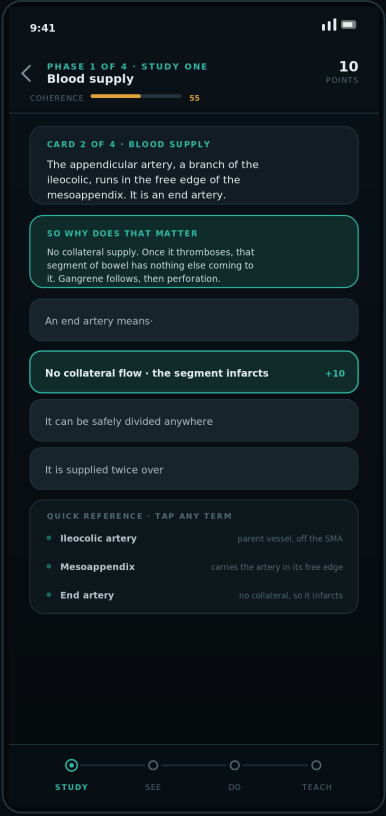
Each card is one paragraph and one question. The paragraph is written the way a good attending explains something at two in the morning, not the way a textbook does.

BUILD Cards are markdown plus a single multiple-choice item. Content lives in a data file so a surgical educator can review and edit it without touching code.

ON SCREEN, TAPPABLE

- Ileocaecal valve
- Lymphoid tissue
- Commensal reservoir

the appendix sits ~2 cm below it
peaks in the second decade
the current best guess at function



04

STUDY ONE

Card 2 · why it dies so fast

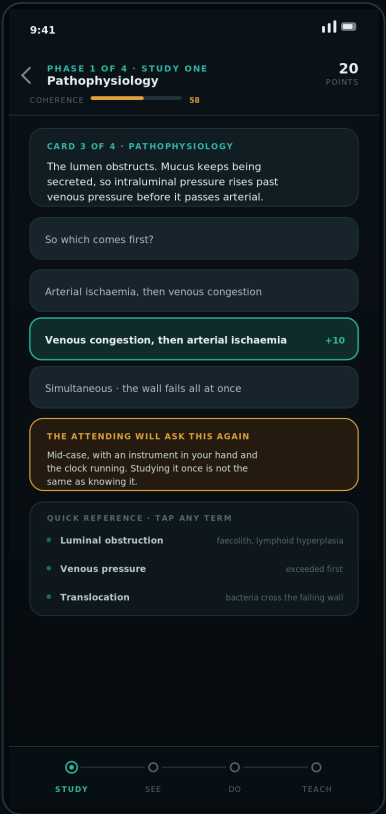
This card is doing double duty. It is the reason a delayed appendicectomy perforates, and it is also the reason the player will have to name the mesoappendix before they are allowed to divide it two phases from now.

BUILD Study content is linked to the operative steps it justifies. Missing a card sends the player back to that specific card at the debrief, not to the whole deck.

ON SCREEN, TAPPABLE

- Ileocolic artery
- Mesoappendix
- End artery

parent vessel, off the SMA
carries the artery in its free edge
no collateral, so it infarcts



05

STUDY ONE

Card 3 · the order of failure

The single most examinable fact about appendicitis, and the one students reliably invert. Getting it right here is worth ten points. Getting it wrong later, when the attending asks it again, costs more than that.

BUILD Flag every study item as re-askable. The mid-case question bank draws from the cards the player has already answered, weighted toward the ones they hesitated on.

ON SCREEN, TAPPABLE

- Luminal obstruction
- Venous pressure
- Translocation

faecolith, lymphoid hyperplasia
exceeded first
bacteria cross the failing wall



06

STUDY ONE

Card 4 · why the pain moves

This is the card that connects the reading back to the man in the first screen. His pain moved at hour nine. The player now knows what that meant, and knew it before they were told the diagnosis.

BUILD Where possible, the study question uses the actual case seed rather than a generic stem. Same content, but the player has to apply it to the patient in front of them.

ON SCREEN, TAPPABLE

- T10 dermatome
- Visceral afferent
- Parietal peritoneum

the umbilicus
poorly localised, midline
somatic, precisely localised



07

THE GATE

Now you may scrub

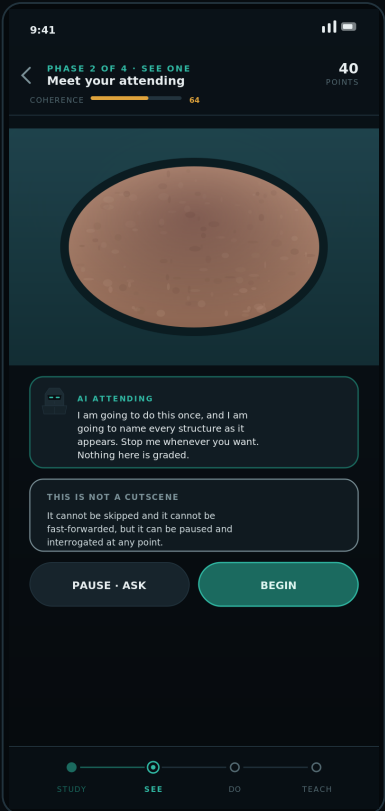
Forty points and the door opens. It is a deliberately small number next to what the operating phases pay, and that asymmetry is the point: studying is the cheapest points in the game and the only ones you cannot skip.

BUILD Score events are a typed ledger, not a running integer. The debrief reconstructs the case from the ledger, which is also what makes head-to-head comparison possible.

ON SCREEN, TAPPABLE

- Points banked
- Score ledger
- Case seed

study score survives a failed case
every event typed and timestamped
makes two runs comparable



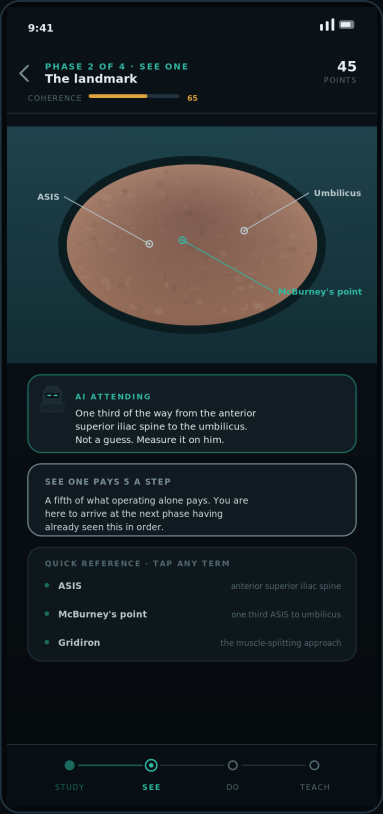
08

SEE ONE

The attending

The attending is armoured, deliberate, and worth impressing. It is the character the whole product is named for, and it must be an original design – a recognisable likeness of an existing film robot is avoidable exposure and would follow the product into every funding conversation.

BUILD Concept art first, before any modelling. The silhouette carries the weight. Voice is synthesised from a written script so content review happens on text.



09

SEE ONE

Where the incision goes

Before a blade appears, the player is shown that the incision is a measurement, not an instinct. The two bony landmarks are labelled on the patient, and the point is derived in front of them.

BUILD Labels are anchored to named vertices on the sourced mesh, not to screen coordinates, so they survive camera moves and work identically in a headset later.

ON SCREEN, TAPPABLE

- ASIS: anterior superior iliac spine
- McBurney's point: one third ASIS to umbilicus
- Gridiron: the muscle-splitting approach



10

SEE ONE

Through the layers, named

The attending cuts and narrates simultaneously. Nothing appears on screen without a name attached, which is the content standard the whole product is built on and the part that cannot be generated loosely.

BUILD Every structure carries three fields: name, what it is, why it matters. If a structure has no clinical line, it does not ship. This is the reviewable artefact.

ON SCREEN, TAPPABLE

- Camper's fascia: fatty superficial layer
- Scarpa's fascia: membranous deep layer
- Aponeurosis: flat tendon of external oblique



11

SEE ONE

The player interrupts

This is the moment See One earns its place. The player stops the attending mid-cut, taps a structure, and gets a real answer – then a hook forward to the step that will make it matter.

BUILD Pause is a first-class state, not an overlay. The clock, the score, and the narration all suspend. Time spent in the encyclopaedia is not penalised anywhere.

ON SCREEN, TAPPABLE

- **Fibre direction** downward and medial
- **Inguinal ligament** the folded lower border
- **Tap to learn** unpenalised, unlimited



12

SEE ONE

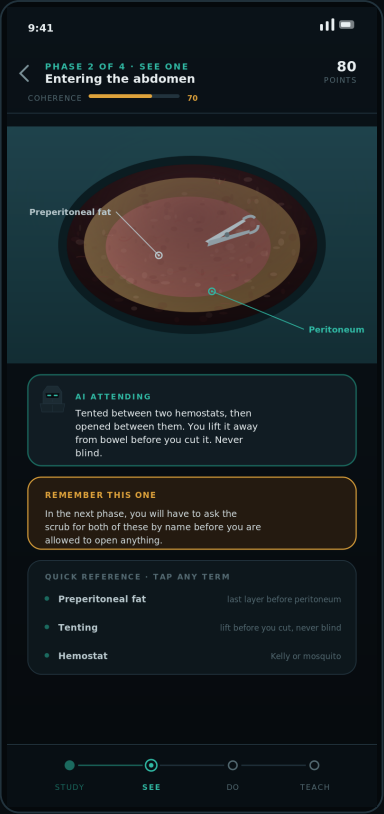
Splitting, not cutting

The single most satisfying idea in the whole operation, and one most students never have explained to them properly: the muscle-splitting incision is strong precisely because it is inconvenient.

BUILD Layer opening is animated per structure with the fibre direction visible. The same asset is reused in Do One, where the player performs the split with a two-finger spread and the tolerance is checked against the fibre axis.

ON SCREEN, TAPPABLE

- **External oblique** down and medial
- **Internal oblique** up and medial
- **Transversus abdominis** transverse



13

SEE ONE

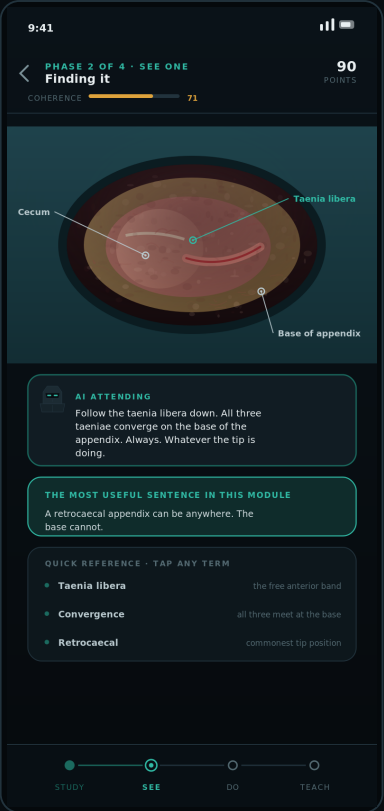
Two hemostats and the peritoneum

A small technical detail that carries real operative fluency. The player is being shown the manoeuvre and simultaneously told that they will shortly be asked to produce the instrument names from memory.

BUILD See One and Do One run off the same step list. A See One step is the same data object with the input layer disabled and the narration enabled.

ON SCREEN, TAPPABLE

- Preperitoneal fat last layer before peritoneum
- Tenting lift before you cut, never blind
- Hemostat Kelly or mosquito



14

SEE ONE

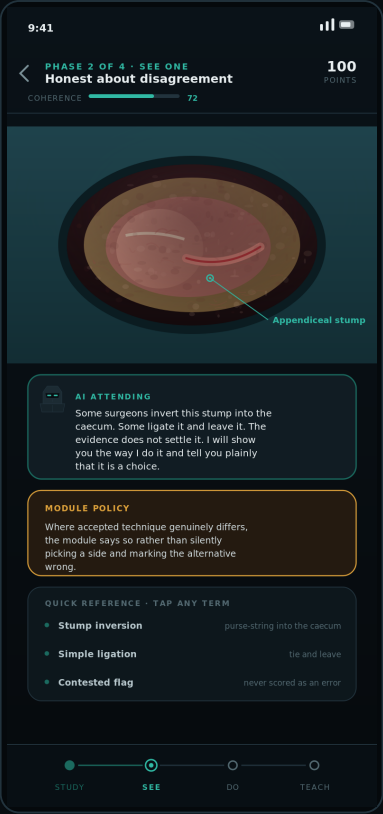
The taeniae converge

This is the fact the player will later be asked to teach a junior, and the reason Teach One works: the best answer in the game is a landmark, not a step.

BUILD Tag structures that appear in the Teach One question bank. The teaching phase should draw on things the player was explicitly taught, never on trivia they were not.

ON SCREEN, TAPPABLE

- Taenia libera the free anterior band
- Convergence all three meet at the base
- Retrocaecal commonest tip position



15

SEE ONE

Where the technique is contested

No competitor does this, and any surgeon reviewing the content will notice immediately that it is here. Admitting the limits of the evidence is what makes the rest of the content credible.

BUILD Steps carry a contested flag. A contested step is never scored as an error either way, and the debrief shows both techniques with the reasoning for each.

ON SCREEN, TAPPABLE

- Stump inversion
- Simple ligation
- Contested flag

purple-string into the caecum
tie and leave
never scored as an error



16

SEE ONE

A question with no warning

The mechanic that converts a demonstration into an examination. It is also, notably, the cheapest thing in the entire design to build – a question, a timer, and a score event.

BUILD Interrupts fire on a step-count trigger with jitter, so a repeat player cannot memorise when they are coming. Drawn from the study cards already banked.



17

DO ONE

The handover

The difficulty does not change between sessions, it changes inside this one. A player who is ahead of the module feels it within thirty seconds rather than at the next unlock.

BUILD Coherence is a rolling function of accuracy and hesitation. It drives three independent knobs – label visibility, gesture tolerance, narration – so it degrades gracefully instead of flipping a difficulty setting.

ON SCREEN, TAPPABLE

- Coherence 76
- Scaffolding
- Ask the attending

above the 72 gate

labels, tolerance, narration

always available, costs nothing



18

DO ONE

You have no instruments

Nearly free to build and one of the highest-value mechanics in the design. It teaches the vocabulary and the social choreography of an operating room at the same time, which is the thing students are actually frightened of.

BUILD Tray contents per step: the correct instrument, two plausible near-misses, and three distractors from elsewhere in the case. Speech recognition is an enhancement, not a dependency – tap always works.

ON SCREEN, TAPPABLE

- Metzenbaum
- Babcock
- Army-Navy

sharp dissection of fascia

atraumatic, for bowel

wound-edge retraction

9:41

105

POINTS

<

PHASE 3 OF 4 · DO ONE

The scrub corrects you

COHERENCE

71

SCRUB TECH

That is a Kelly. You asked me to cut fascia with a clamp. Do you want the Metzenbaums?

WRONG INSTRUMENT

-10

Case continues · coherence -6

NOBODY HANDS YOU THE WRONG THING SILENTLY

The error is named, the reason is given, and the case carries on. Failure here is instructional, not terminal.

Metzenbaum scissors

sharp dissection

Hemostat

Kelly · you asked for this

QUICK REFERENCE · TAP ANY TERM

Wrong structure

you did not know it

Wrong manoeuvre

you knew it, wrong action

Wrong place

right action, imprecise

●

●

●

○

STUDY

SEE

DO

TEACH

19

DO ONE

Wrong instrument

The failure branch matters more than the success branch. A wrong answer produces a specific human correction rather than a buzzer, which is exactly how it happens in a real operating room.

BUILD Three distinct error classes, scored separately: wrong structure, wrong manoeuvre, and right manoeuvre in the wrong place. They are three different kinds of not knowing and the debrief must distinguish them.

ON SCREEN, TAPPABLE

- **Wrong structure** you did not know it
- **Wrong manoeuvre** you knew it, wrong action
- **Wrong place** right action, imprecise



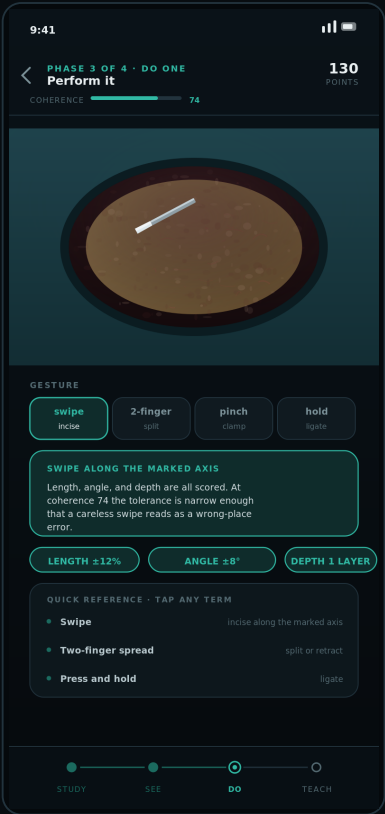
20 DO ONE

Name it before you cut it

The signature mechanic

This is the sentence the entire product is built around, and it is also the claim that has to be tested first: does identify-before-you-cut teach anatomy better than a labelled diagram? Forty students and a pre-post quiz would answer it.

BUILD Gesture input is disabled until the identification resolves. No timeout on the first attempt in entry modules – the timer only appears once the player clears coherence 72.



21

DO ONE

Then the gesture

Four gestures cover the entire operative vocabulary: swipe to incise, two-finger spread to split or retract, pinch to clamp, press-and-hold to ligate. A phone can do all four without an install.

BUILD Gesture tolerance is a single coherence-driven scalar applied to length, angle, and depth thresholds. The same gesture map becomes trigger-pull semantics in a headset and physical actuation with the Bluetooth instruments later.

ON SCREEN, TAPPABLE

- **Swipe** incise along the marked axis
- **Two-finger spread** split or retract
- **Press and hold** ligate



22

DO ONE

Right manoeuvre, wrong place

Most simulators have one failure state. This one has three, and the distinction is what lets the debrief tell a player whether they have an anatomy problem or a technique problem.

BUILD Error taxonomy is fixed at three classes and never collapsed for convenience. The remediation path differs: wrong structure sends you back to a card, wrong place sends you back to a gesture drill.

ON SCREEN, TAPPABLE

- **Rectus sheath** medial to the gridiron
- **Linea semilunaris** the lateral border of rectus
- **Remediation** gesture drill, not a card

9:41

<

PHASE 3 OF 4 · DO ONE

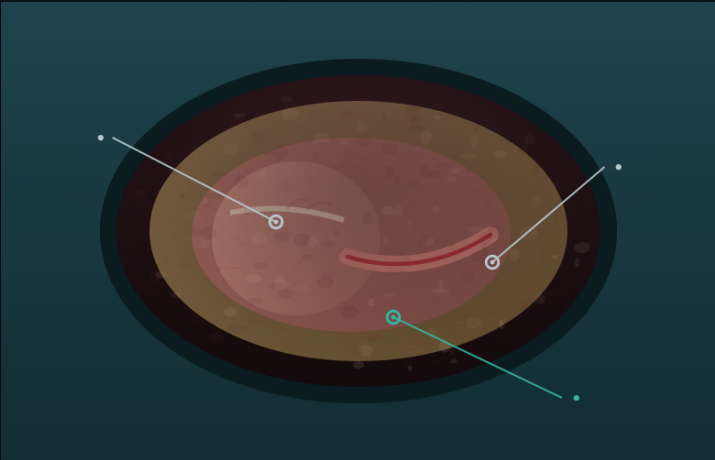
180

The decision

POINTS

COHERENCE

78



 AI ATTENDING

What are you about to divide? Fourteen seconds.

Mesoappendix, carrying the appendicular artery

+25

Taenia libera

-10

Ileocolic artery

-25

Peritoneal reflection

-10

LABELS ARE GONE

At coherence 78 the structures are unlabelled dots. You are naming them from the study cards and from See One, not reading them off the screen.

STUDYSEE**DO**TEACH

23 DO ONE

What are you about to divide?

The beat that answers “what do I actually do?”

The beat that answers the question a player asks about this product in the first thirty seconds: what do I actually do? You are asked what to do next, and you have to know.

BUILD The distractor set is generated from structures genuinely present in the field at that step, never from a global list. A plausible wrong answer is what makes the right one worth something.



24 DO ONE

Press and hold to ligate

The study card, the See One narration, and the manoeuvre finally close on each other. Nothing in the reading was decoration – every card exists because a later step depends on it.

BUILD Every study card declares which operative steps it justifies. If a card has no downstream step, it is trivia and should be cut from the deck.

ON SCREEN, TAPPABLE

- Proximal ties two, then divide
- Retraction risk the stump disappears into mesentery
- Card 2 this is where it gets spent



25 DO ONE

Coherence falls

The direction most difficulty systems never go. Coherence is not a score to be beaten – it is a thermostat, and it is the mechanic that decides whether a nervous student finishes the case or closes the app.

BUILD Hesitation is measured as time-to-first-input relative to that player's own rolling baseline, not an absolute threshold. Otherwise you punish careful readers and reward guessers.

ON SCREEN, TAPPABLE

- Hesitation measured against your own baseline
- Coherence 58 between the 34 and 72 gates
- Repeat Do One the outcome below 72

9:41

PHASE 3 OF 4 · DO ONE

Closure

280
POINTS

COHERENCE

82

AI ATTENDING

Name them going out in the order you met them going in. Peritoneum, transversus, internal oblique, external oblique aponeurosis, Scarpa's, skin.

Close the aponeurosis with:

0-Vicryl, continuous

+25

3-0 nylon, interrupted

-10

Skin staples

-10

STUDY

SEE

DO

TEACH

Closing, in reverse

Closure is where most simulators stop bothering. Here it is the recall test: the same six layers, in reverse, from memory, with the suture choice scored.

BUILD The closing sequence is the opening step list reversed, so it costs almost nothing to author and doubles as a free retention check on the layer anatomy.

9:41

<

PHASE 4 OF 4 · TEACH ONE

280

You are the senior now

POINTS

COHERENCE

84

JUNIOR · THIRD-YEAR STUDENT

I cannot find the appendix. The tip is not where I expected it. What is the landmark?

WORTH 40

HIGHEST IN THE GAME

WHY THIS IS THE MOST VALUABLE PHASE

Being able to perform a procedure and being able to be responsible for someone else performing it are different competencies. Almost nothing on the market tests the second one.

TEACH ONE · THIS CASE

Q1 · the landmark for the appendix

40

Q2 · why the incision is oblique

40

Q3 · trace that artery to its source

40

TYPE

SPEAK

QUICK REFERENCE · TAP ANY TERM

Teach One

40 a step, the highest in the game

Rubric

concepts, not keywords

Free text

typed or spoken

STUDYSEEDO

TEACH

27 TEACH ONE

A junior is asking

The phase nothing else on the market tests

The differentiated mechanic and the untested one. Whether Teach One retains people is one of the three things this product has to prove before anyone should fund it.

BUILD Free-text answer graded against a rubric of required concepts, not string matching. This is the one place a language model earns its keep – and the one place its output must be reviewable by an educator.

ON SCREEN, TAPPABLE

- Teach One40 a step, the highest in the game
- Rubricconcepts, not keywords
- Free texttyped or spoken



28 TEACH ONE

Your answer is scored

Partial credit against named concepts, so the player can see exactly which part of the explanation they left out. The missing concept becomes the next question.

BUILD Rubric items are authored alongside the study cards by the same reviewer. The model checks for concept presence and returns which items were hit – it does not invent the rubric.

ON SCREEN, TAPPABLE

- Partial credit shows you what you left out
- Retrocaecal variation the item you missed
- Educator review rubrics authored with the cards



29 TEACH ONE

The harder question

This is the question a real third-year would ask, and the one that separates a memorised answer from an understood one. There is no hint button on this screen.

BUILD Follow-up selection is deterministic given the rubric result, so two players who give the same answer get the same follow-up and the comparison stays fair.

ON SCREEN, TAPPABLE

- Adaptive follow-up targets the missing concept
- Deterministic same answer, same follow-up
- No hint this is the competence test



30 DEBRIEF

The case, reconstructed

The debrief is rebuilt from the score ledger, so it can name the specific card behind a specific hesitation. That traceability is what makes a bad outcome a teaching moment rather than a punishment.

BUILD Never show a total without its derivation. Every line is a query against the event ledger, which is also the export a school would want for assessment.

ON SCREEN, TAPPABLE

- Traceability every total shows its derivation
- Remediation target the specific card, not the deck
- Export the ledger is the assessment record



31 THE LADDER

Where the arguments start

This is where two students end up arguing in a corridor about which clamp belongs on an aortic arch, and are both able to say why. That argument is the product working. Everything before it is the setup.

BUILD Modules are pure data – layers, structures, brief, steps, tray – on one shared engine. The engine ships once. After that a new module is a content job, which is what makes the ladder affordable.

ON SCREEN, TAPPABLE

- Critical view of safety the next module's whole point
- Trauma mode randomised, clocked, partial view
- Modules are data the engine ships once

What this actually is, as a build

The expensive-looking part of this product is not the expensive part. Rendering an operating room to a standard that competes with a funded studio is out of reach and is not the plan. The attending asking a question, scoring the answer, and adapting the difficulty is text, a timer, and a state machine — and it is the entire differentiator. Anatomy is bought in, not modelled. Everything else is a small team writing an engine once and then authoring content.

THE PIECES, IN THE ORDER THEY SHOULD BE BUILT

- 1 The step engine** A module is data: layers, structures, brief, steps, tray. See One and Do One are the same step list with the input layer toggled. Build this once.
- 2 The identify-to-unlock gate** Closures generated from structures genuinely present at that step. This is the signature mechanic and it is a week of work, not a quarter.
- 3 The score ledger** Typed, timestamped events. Everything else — debrief, head-to-head, school reporting — is a query against it.
- 4 Coherence** A rolling function of accuracy and hesitation driving three knobs: labels, gesture tolerance, narration. Hesitation is relative to the player's own baseline.
- 5 Sourced anatomy** Import and tag, do not model. Licensing on the share-alike datasets needs counsel before commercial release — that is a real open question, not a formality.
- 6 Teach One** Free-text answers against an educator-authored rubric. The one place a language model earns its keep, and the one place its output must be reviewable.

WHAT HAS TO BE PROVEN FIRST

Does identify-before-you-cut teach anatomy better than a labelled diagram? Pre/post quiz, forty students, costs almost nothing. Does Teach One retain people? Will a named surgical educator sign their name to the content? One yes to the third is worth more than any market-size slide.

WHAT IT COSTS, HONESTLY

Four to six modules on a phone is a small-team build: one engine developer, one 3D anatomy artist working from sourced data, and a surgical content reviewer. Nine to twelve months. Low seven figures at the top end and considerably less scrappily. Nothing here requires a nine-figure raise.

Two students arguing in a corridor about which clamp belongs on an arch, and both able to say why.

That is the product working. Everything in this document is the setup.

CONTACT

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Twenty-plus years in the operating room. The clinical content, the case ladder, and the design in this document are mine. The engineering is not, and that is the conversation I am trying to have.

WHAT I AM LOOKING FOR

A technical partner or a small studio to build Module 01 end to end — the step engine, the identify-to-unlock gate, and the score ledger — as a playable vertical slice on a phone. One module, one case, finished. Not a prototype of everything. Read the beats. If the gate on page 4 and the phase on page 19 look buildable to you, that is the whole conversation.

A NOTE ON THIS DOCUMENT

This storyboard follows the August 2026 scoped concept. It replaces the VR framing in the earlier master business plan — this is a phone product, and VR is a later phase of the same codebase rather than a separate thing to raise against.

Screen designs are final-intent. The anatomy rendered in them is placeholder and must be replaced with sourced geometry before this is shown to a surgical audience. Market figures from the original business plan are deliberately absent here; several of them need re-verification against primary sources or removal before any version of this goes to an outside reader. Nothing in this document depends on them.